



Making the BESSt out of green energy

Drake Consulting

How should European Energy expand its presence in the BESS market in 2026-2028, creating long-term value while managing risk and capital limitations?

- Analysis:**
- 1 Situation Analysis**
Market volatility, limited capital, need for profitability
 - 2 Market Expansion Analysis**
Country prioritization, contract mix, innovation roadmap



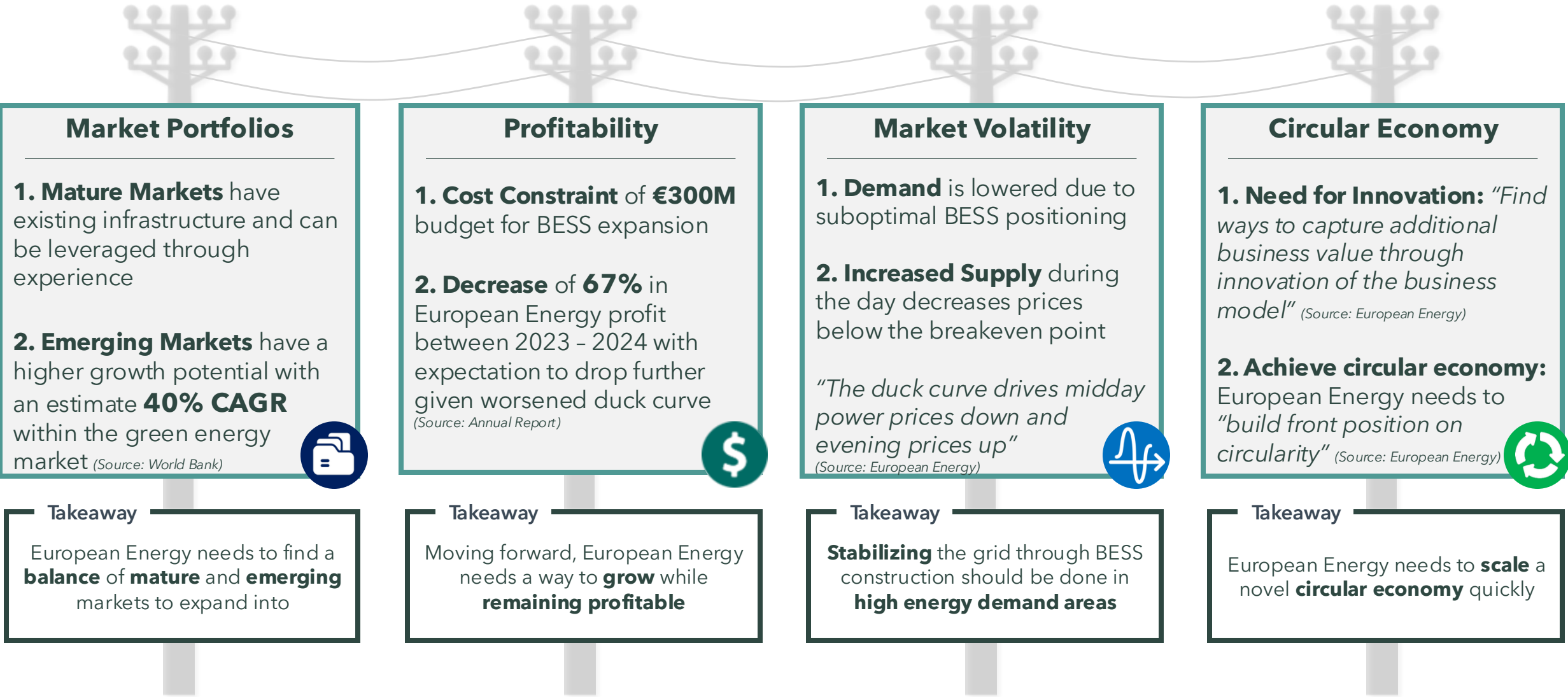
330
Million Euros in
Annual Revenue

5
Key Projects

3.08
GW of Power
Deployed

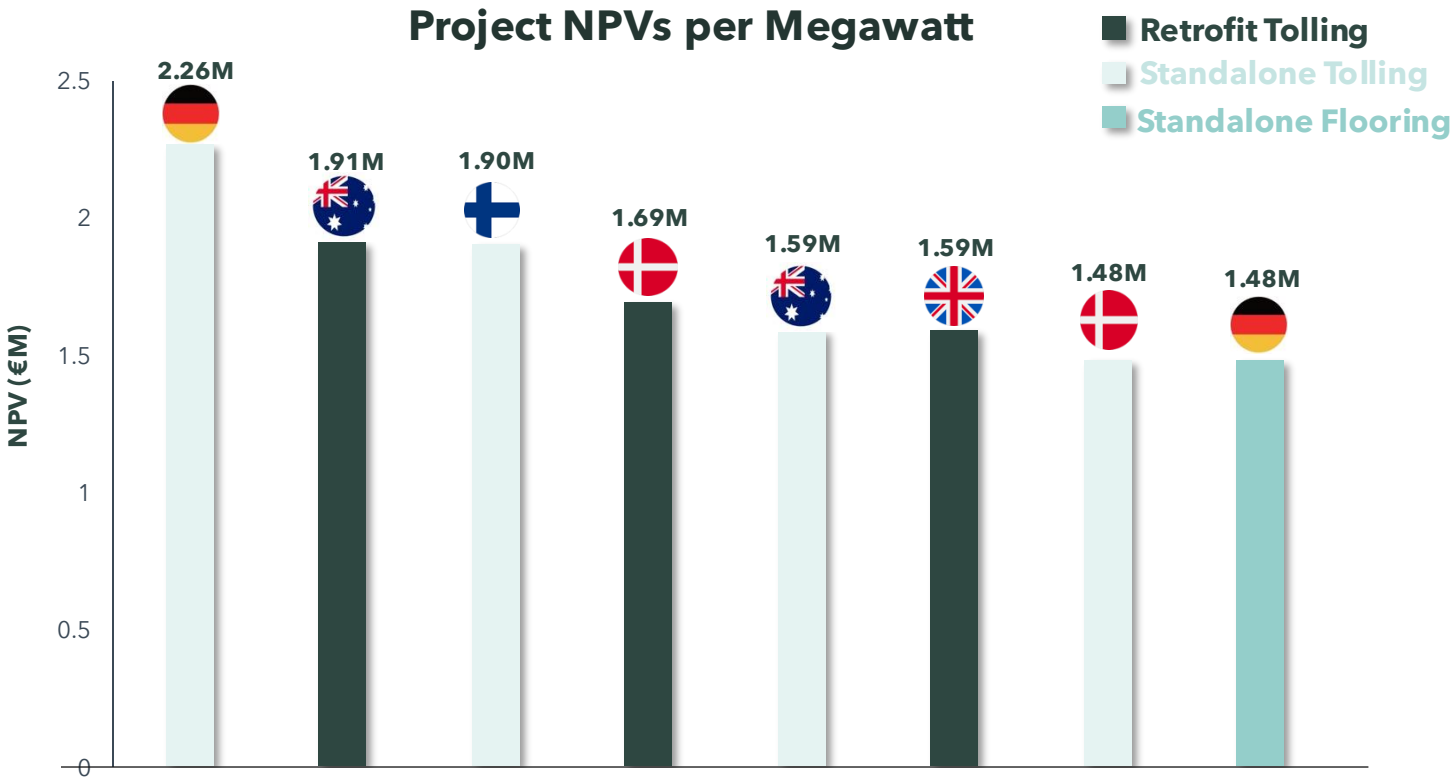
77%
Retrofitting
Projects





European Energy's BESS investment should focus only on tolling offtake agreements

Market Analysis



The graph shows the eight highest NPV per MW projects out of all countries, offtake agreements and BESS types

Calculation = Present Value of Annual Revenue Stream - Present Value of Annual OpEx - CapEx

Key Takeaways

Offtake Agreement

When evaluating the **highest NPV projects**, the tolling offtake agreement is used throughout the top 7

Based on this data, it is clear European Energy should **strictly pursue tolling** offtake agreements to **increase profitability**

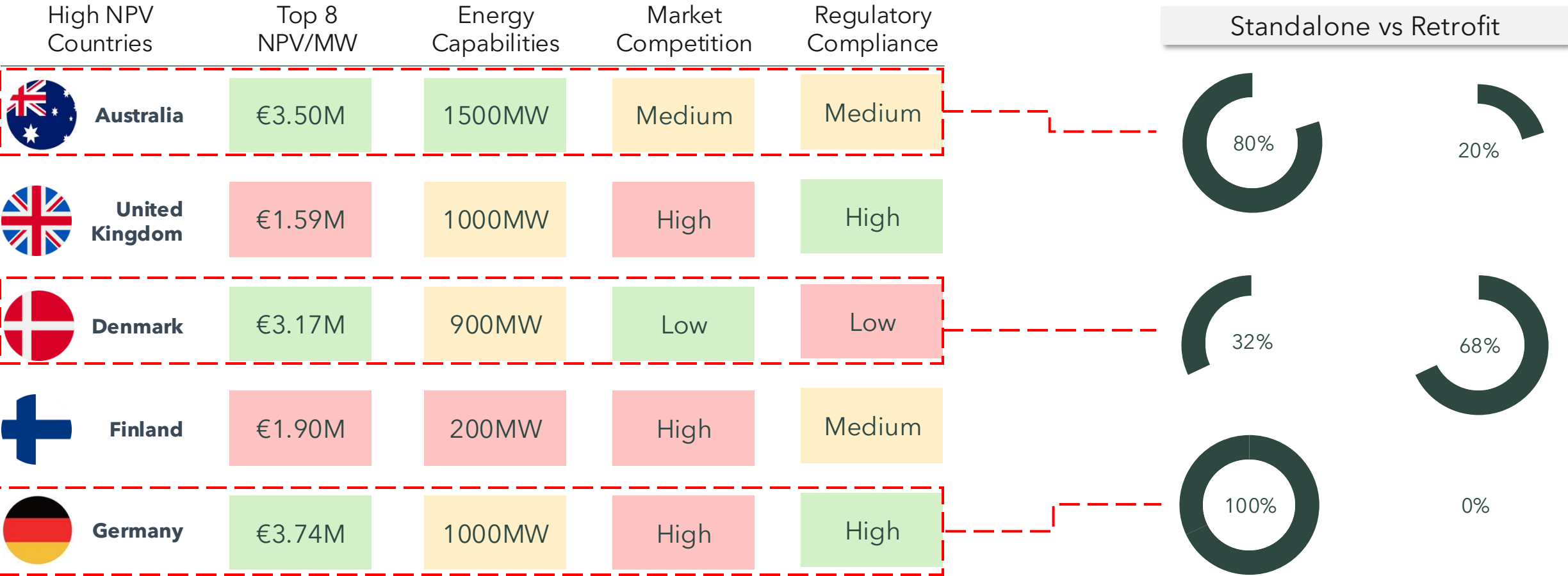
Key Countries

After evaluating all 10 proposed countries, **Germany, Australia, and Denmark** appear to be the countries that have the **highest NPV** investments with each country having two **different** projects within the top 8



Australia, Denmark, and Germany are key markets for 2026-2028 BESS investment

Market Analysis



Based on the key metrics above, **Australia, Denmark and Germany** show the best opportunities for expansion

Key Principles addressed:


Market Portfolio


Profitability


Volatility



Project BESSt

1

Smart Grid Expansion

Build standalone and retrofit BESS near AI data centres and renewables to stabilize high-load grids

2

Circular Battery Loop

Recycle EV and old BESS batteries into second-life systems to cut costs, reduce emissions, and secure long-term material supply

Project BESSt

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Guiding question:

Where should European Energy build BESS to **stabilize high-load grids** and **meet 24/7** clean-energy demand?

An exciting new market: AI data centers

- 1 Global data-centre electricity use will **more than double** by 2030
(Source: IEA)
- 2 Frankfurt's data centres alone consume **~40%** of local electricity
(Source: McKinsey)
- 3 Most AI data centres **lack on-site generation**, relying entirely on strained local grids
- 4 Microsoft, Google, Amazon, and Meta all target **24/7 carbon-free energy** for their AI infrastructure

7.2% CAGR
2026-2030

Why **standalone BESS** near AI data centres?



Standalone batteries near high-demand clusters stabilize the grid and **maintain 24/7 reliability**



Ideal for data centres without on-site generation, ensuring **consistent** power and **reduced** grid stress

Why **retrofit BESS** near solar and wind farms?



Overcome duck curve loss



Leverage off existing infrastructure



Provide grid stability

Key Principles addressed:



Market Portfolio



Profitability



Volatility



Circular Economy



- 1

Identify & target high-load nodes
Pinpoint substations near AI data-centre clusters and renewable hubs with major grid congestion in Germany, Denmark, and Australia
- 2

Partner with TSOs / DSOs for pilot projects
Launch multi-site pilots per country co-developed with grid operators to provide balancing, peak-shaving, and congestion relief



R

Underperformance of pilots or technical integration issues

M

Start with small-scale modular units (10-20 MW) and monitor KPIs jointly with grid operators to refine performance

- 1

Scaling through visibility and replication
Publicize operational success through energy conferences, media, and policy networks and use performance data to replicate standardized “grid-support storage” contracts

2026
Phase 1:

2027
Phase 2:

2028-29
Phase 3:



R

Slow partner buy-in or regulatory delays

M

Engage early with grid operators through joint feasibility studies and align with national storage targets

- 1

Execution: pilot and proof of impact
Deploy multi-site BESS pilots in each country to deliver balancing, peak-shaving, and congestion-relief services



R

Limited visibility or policy traction

M

Leverage case studies and cost-saving evidence to lobby ministries and investor forums; publish verified impact metrics

Key Takeaway

European Energy scales by proving grid value first. Partnerships, performance, and policy drive adoption faster than marketing

Key Principles addressed:

Market Portfolio

Profitability

Volatility

Circular Economy

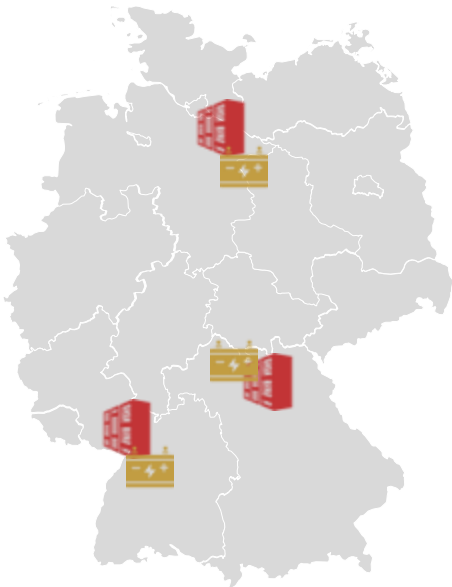


-  **Data Centre**
-  **Standalone**
-  **Retrofit**

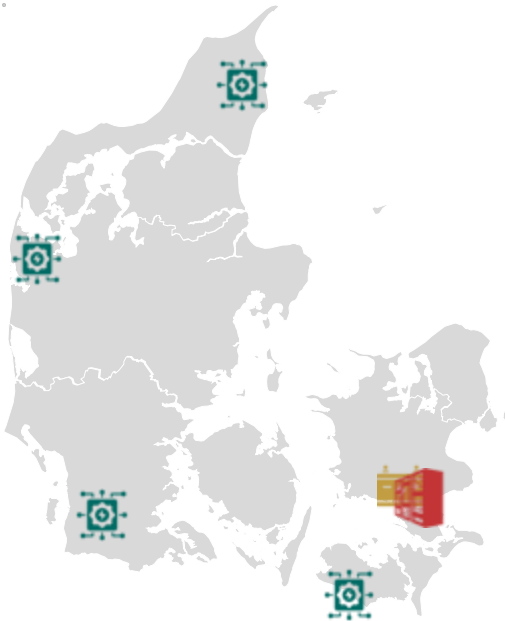
Strategic Deployment Focus

Prioritizing standalone builds near data-centre grids and retrofits near renewables to **maximize grid stability and profitability**. Targeting data-centre corridors and renewable hubs across key markets for optimal grid impact

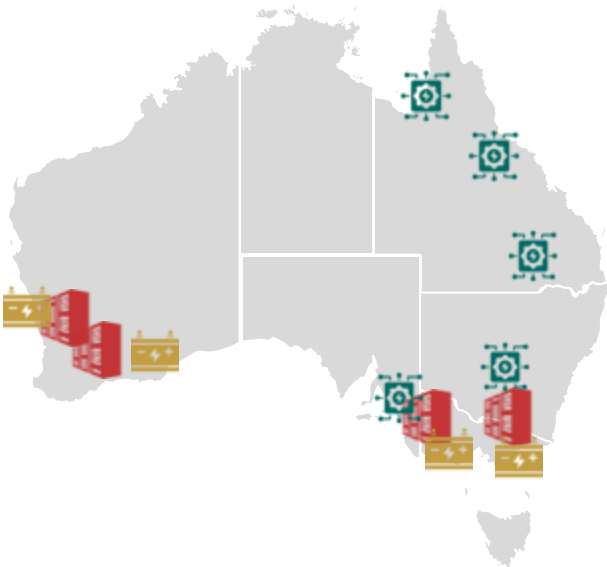
Germany



Denmark



Australia



Project BESSt

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Smart Grid Expansion

Build standalone and retrofit BESS near AI data centres and renewables to stabilize high-load grids

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Circular Battery Loop

Recycle EV and old BESS batteries into second-life systems to cut costs, reduce emissions, and secure long-term material supply

Project BESSt

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Smart Grid Expansion

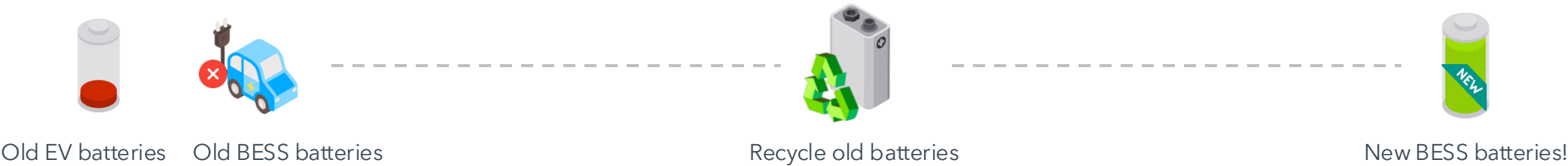
Build standalone and retrofit BESS near AI data centres and renewables to stabilize high-load grids

2

Circular Battery Loop

Recycle EV and old BESS batteries into second-life systems to cut costs, reduce emissions, and secure long-term material supply

Lithium Battery Recycling is a circular battery ecosystem that recycle EV batteries and old BESS units to create new BESS



Why battery recycling?

Achieve a **circular economy** and address **sustainability** by reusing valuable resources



Decrease in battery life span after being recycled

1. Cost Savings

Decreased lifespan is outweighed by increase in cost savings
Second-life modules can cost **<60%** of equivalent new battery packs
(Source: McKinsey)



2. Faster Production

Decrease time to deploy while maintaining efficacy
20% decrease in time to deploy *(Source: PwC)*

3. Sustainability



Align with company value:
"Become the global **leader** in the **green transition**" *(EE vision statement)*

Recycling lithium-ion batteries can yield **~72-88%** less water use, **~77-89%** less energy compared to mining virgin materials *(Source: Stanford University)*

4. Overcome Natural Resource Shortage



Address shortage of key materials used in battery production

Global supply will fall about **50%** short of demand by **2030** *(Source: International Energy Agency)*



How does it work?

European Energy use renewed EV packs from automakers, use them in stationary BESS, and recycle them for metal recovery

Step 1:
Locate **TWO** types of vendors to:

- 1. Purchase old batteries
- 2. Rebuild new BESS

1. Automobile companies
To purchase old EVs



Potential Automobile Vendors:
The Volvo Group
Renault

2. Battery recycling companies
To rebuild old batteries into new BESS



Potential EV Recycling Vendors:
The Volvo Group
NorthVolt, Connected Energy:
European based company, strong record of EV battery recycling and expertise in BESS recycling

Step 2:
European Energy will sign framework agreements with partners to secure supply, follow compliance, and have quality and safety controls

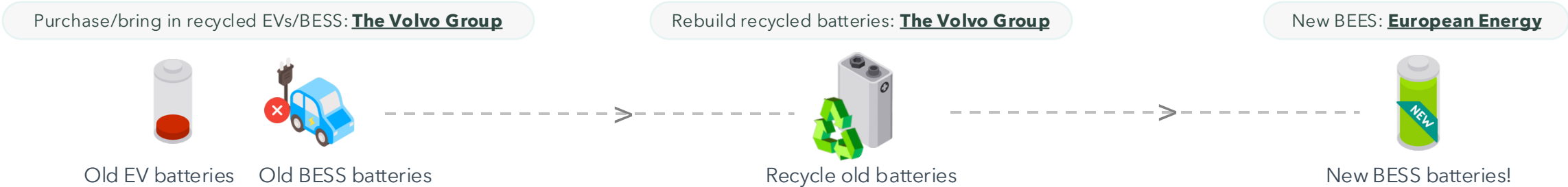
Highest Potential Vendor

- 1** **The Volvo Group**, an automotive manufacturing company that **supplies** and **recycles** old EV batteries
- Large market share of EV in **Europe**
 - Provides **battery passport** & **battery history** (required for EV reproduction by law)



Recycling batteries allows European Energy to establish a successful circular economy through business model innovation and proven success

Solution 2: Circular Battery Loop



We know this solution will work because...

Case Study

Rome Airport's partnership with ENEL used government funding to build recycled EV BESS as a backup energy storage systems. With proven strategies used within the European market, European Energy can model after similar projects

Case Study

Redwood Materials was able to leverage nearby data centers to ensure a successful use of recycled EV batteries provided from General Motors. With it being the second largest second-life battery deployment, European Energy has an opportunity to set a new global standard for battery recycling with proven success to model after

Key Principles addressed:

Market Portfolio

Profitability

Volatility

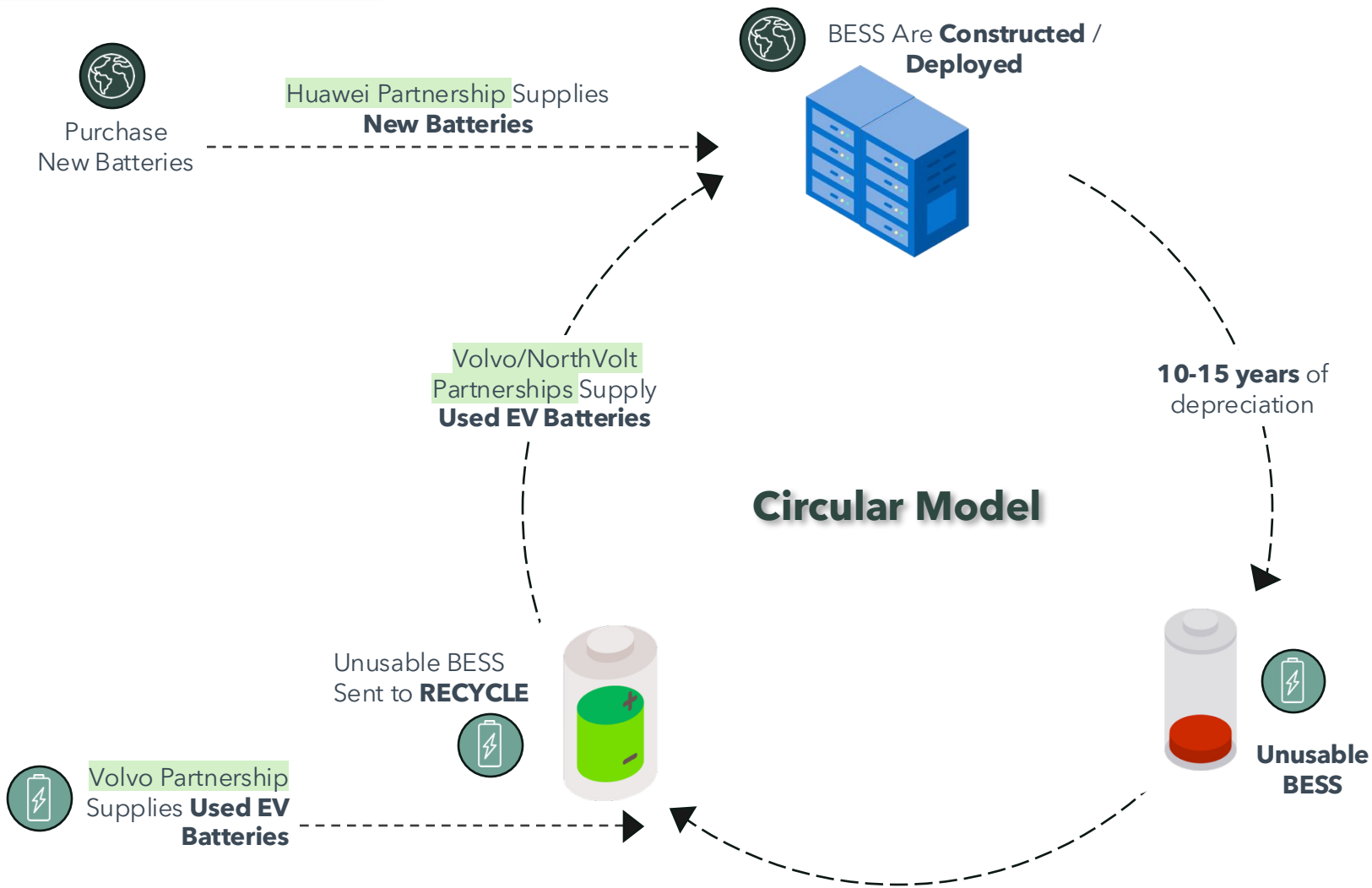
Circular Economy

European Energy can rely on recycled EV and BESS batteries given their proven success in diverse markets

European Energy should start tomorrow on identifying sites and securing partnerships

Solution Summary

Recap of project BESSt



Legend



Solution 1 (Smart grid expansion)



Solution 2 (Circular battery loop)

Highlight

Vendor Partnerships

Tomorrow's Tasks

1. Identify sites near existing projects and grid-powered AI data centres
2. Contact DSO&TSO to inform them of addition to local grid supply
3. Contact new and old battery supply companies as well as recycling companies to aid with faster integration (ex. Volvo Group)

Key Principles addressed:



Market Portfolio



Profitability



Volatility

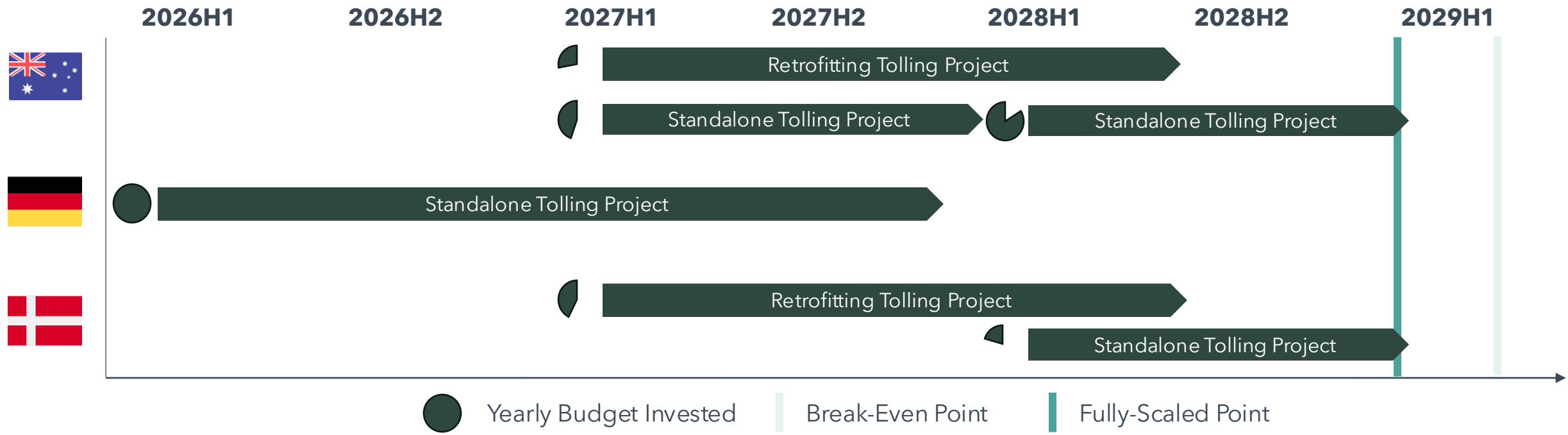


Circular Economy



European Energy will utilize their €300 Million budget on projects by 2029

Implementation Timeline



Risks and Mitigations

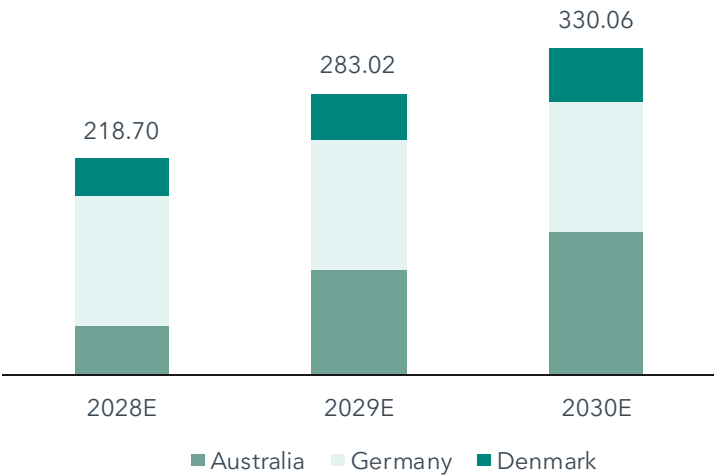
Most Severe ↑ ↓ Least Severe	1	Limited Value-Chain Ownership	➤	Maintain operational agility, allowing to capture margins, improve responsiveness, and competitive positioning
	2	Higher WACC from Elevated Debt Levels	➤	Utilize high-performing assets as collateral to negotiate more favorable financing terms and optimize capital structure
	3	Issues on Divesting New Projects	➤	Adopt a merchant ownership model to monetize assets more effectively and diversify revenue streams

European Energy deploys over 3.08 GW of power and earns €612.72M by 2030

Financial Impact

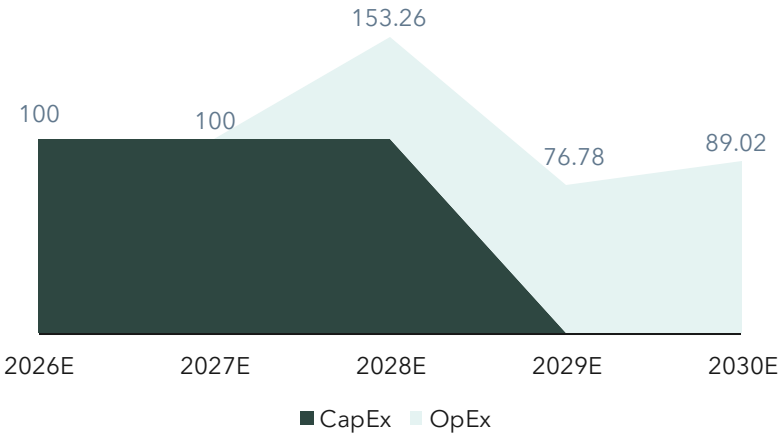
Incremental Revenue

(in millions €)



Costs

(in millions €)

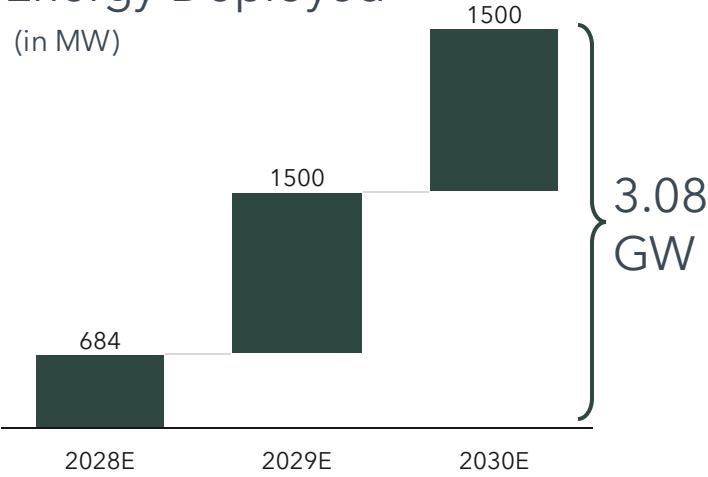


Key Assumptions

- 1 First revenue generated in 2029, due to 2-year development period
- 2 Focusing on highest NPV per Megawatt projects
- 3 Project revenue decreases 2% per year, while WACC remains constant

Energy Deployed

(in MW)



Key Statistics



Retrofitting
vs
Standalone



Growth
vs
Mature

Net Present Value (NPV)



2 Projects - €2.42 Billion



1 Projects - €2.26 Billion

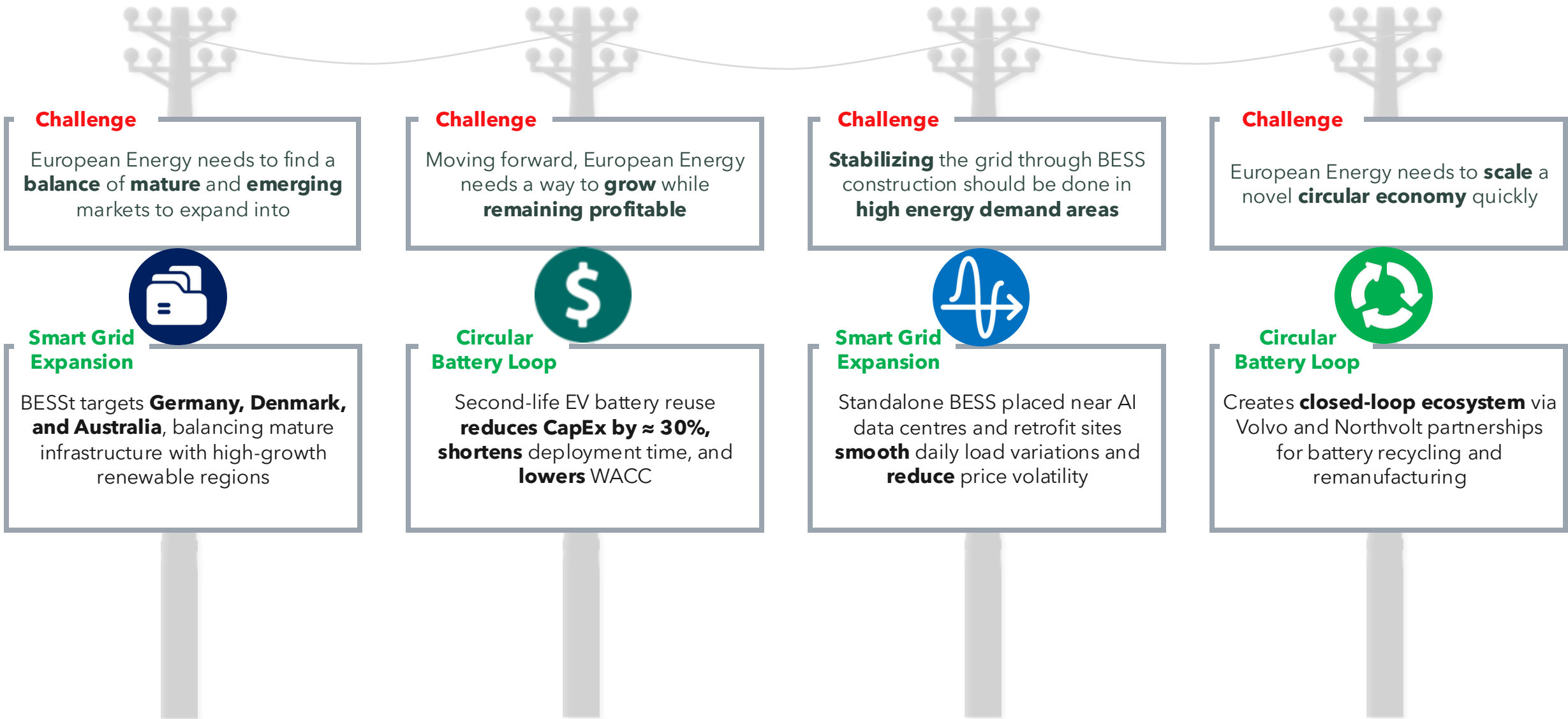


2 Projects - €950.67 Million



Project BESSt directly addresses European Energy's core strategic challenges

Overall Summary





Making the BESSt out of green energy

Drake Consulting

Energy Deployed					
Revenue Projections					
(€ million)	2026E	2027E	2028E	2029E	2030E
Incremental Revenue	-	-	218.70	283.02	330.06
Australia	-	-	50.70	106.74	145.50
Germany	-	-	130.00	130.00	130.00
Denmark	-	-	38.00	46.28	54.56
Operating Costs	-	-	53.26	76.78	89.02
Australia	-	-	13.26	31.26	43.50
Germany	-	-	30.00	30.00	30.00
Denmark	-	-	10.00	15.52	15.52
Incremental Profit					
Australia	-	-	37.44	75.48	102.00
Germany	-	-	100.00	100.00	100.00
Denmark	-	-	28.00	30.76	39.04
Profit	0	0	165.44	206.24	241.04

Debt & Equity			
Equity Schedule			
(€ million)	2026E	2027E	2028E
Australia	-	64.8	81.6
Germany	100.00	-	-
Denmark	-	35.20	18.40
Total	100.00	100.00	100.00
Debt Schedule			
(million €)	2026E	2027E	2028E
Australia	0	259.2	585.6
Germany	400	400	400
Denmark	0	140.8	214.4
Total	400	800	1200

Energy Deployed			
Energy Deployed (MW)	2028E	2029E	2030E
Australia	684	1500	1500
Germany	1000	1000	1000
Denmark	400	584	584
Total	2084	3084	3084

Projects						
Projects (€ million)	Annual Revenue	CAPEX	OPEX	Annual Profit	NPV	Capacity (MW)
Australia	145.50	146.40	43.50	102.00	2479.01	1500
Retrofitting - Tolling	31.50	26.40	7.50	24.00	572.72	300
Standalone - Tolling	114.00	120.00	36.00	78.00	1906.29	1200
Germany	130.00	100.00	30.00	100.00	2264.25	1000
Standalone - Tolling	130.00	100.00	30.00	100.00	2264.25	1000
Denmark	54.56	53.60	15.52	39.04	950.67	584
Retrofitting - Tolling	38.00	35.20	10.00	28.00	677.67	400
Standalone - Tolling	16.56	18.40	5.52	11.04	273.00	184

Project Sensitivity						
(€ million)	Tolling		Flooring		Merchant	
	Retrofit	Standalone	Retrofit	Standalone	Retrofit	Standalone
UK	1.59	1.17	1.04	0.70	0.89	0.56
Poland	N/A	N/A	N/A	N/A	0.89	0.66
Germany	N/A	2.26	N/A	1.48	N/A	1.28
Denmark	1.69	1.48	1.11	0.93	0.96	0.78
Lithuania	N/A	N/A	N/A	N/A	1.07	0.89
Australia	1.91	1.59	1.27	1.01	1.11	0.85
Finland	N/A	1.90	N/A	1.24	N/A	1.06
Czechia	N/A	N/A	N/A	N/A	N/A	0.62
Romania	N/A	N/A	N/A	N/A	N/A	0.59
Greece	N/A	N/A	N/A	N/A	N/A	0.59

